academicthesis-polimi

Politecnico di Milano layouts for the `academicthesis` class

User and maintainer manual

Version 0.1.0 — October 2026

Abstract

`academicthesis-polimi` is the Politecnico di Milano plugin of the modular `academicthesis` class. It reproduces the three official thesis layouts of the School of Industrial and Information Engineering (Scuola 3I, template revision October 2021): the *classical* thesis, the *article* thesis and the *Executive Summary*. You only choose `university=polimi` and a format; the package picks the base class, page layout, colours, cover and headings.

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1. The three formats

Format	What it is	Built on	Showcase
`classical` (default)	The full thesis as a book: cover page, abstracts, chapters	`book`, 12pt, two-sided	`example-polimi-classical.tex`
`article`	The thesis as a paper: header block with abstract on page 1, sections	`article`, 11pt	`example-polimi-article.tex`
`summary`	The Executive Summary: two columns, blue page header	`article`, 11pt, two columns	`example-polimi-summary.tex`

All formats are A4. The showcases, in the `showcases/` folder of the repository, are complete compilable examples: copying one is the fastest way to start.

2. Quick start

```
\documentclass[university=polimi, master, format=classical,
               language=english]{academicthesis}

\thesissetup{
  title         = {Title},
  candidate     = {Name Surname},
  idnumber      = {000000},
  degree        = {Xxxxxxx Engineering - Ingegneria Xxxxxxx},
  supervisor    = {Prof. Name Surname},
  cosupervisors = {Name Surname, Name Surname},
  academicyear  = {2025-26},
}

\begin{document}
\frontpage

\frontmatter
\begin{abstractthesis}[keywords = {first, second, third}]
  English abstract.
\end{abstractthesis}
\begin{abstractthesis}[language = italian,
                       keywords = {primo, secondo, terzo}]
  Abstract in italiano.
\end{abstractthesis}
\tableofcontents
```

```
\mainmatter
\chapter{Introduction}
...
\end{document}
```

3. Class options

All options go in `\documentclass[...]{academicthesis}`.

Option	Values	Notes
`university`	`polimi`	Loads this package.
degree	`bachelor`, `master`	Prints "Bachelor's / Master's Thesis in" (English) or "Tesi di Laurea Triennale / Magistrale in" (Italian). `phd` is not available yet.
`format`	`classical`, `article`, `summary`	Default `classical`. Any other value is an error that names the three formats.
`language`	`english`, `italian`, ...	Main language of the thesis and of every label.
`other-languages`	e.g. `{french}`	Extra languages for blocks written in another language. English and Italian are always loaded for PoliMi, so the second abstract needs nothing here.
`secondpage`		PoliMi has no examination-commission / copyright page, so it is *off* by default; `secondpage` adds the core one.
`11pt`, `oneside`, ...		Standard class options still work and win over the format's defaults.

4. Thesis data (`\thesissetup`)

PoliMi uses the *core names* of the class unchanged; only the printed labels follow PoliMi's wording. A missing required key stops the compilation with a message that names it and shows an example.

Key	PoliMi meaning (printed as)	classical	article	summary
`title`	Title	req.	req.	req.
`subtitle`	Subtitle	opt.	opt.	opt.
`candidate`	Author (Author / Candidato)	req.	req.	req.
`idnumber`	Student ID (Student ID / Matricola)	req.	req.	—
`degree`	Study programme, after “Master’s Thesis in”	req.	req.	req.
`supervisor(s)`	Advisor(s) (Advisor / Relatore)	req.	req.	req.
`cosupervisor(s)`	Co-advisor(s), comma-separated (Co-advisor / Correlatore)	opt.	opt.	opt.
`academicyear`	Academic year (Academic Year / Anno Accademico)	req.	req.	req.
`abstract`	English abstract printed on page 1	—	req.	—
`keywords`	Keywords of that abstract, in a blue box	—	opt.	—
`gender`	`male` / `female` (Candidato / Candidata)	opt.	opt.	opt.

5. PoliMi options (`\polimithesissetup`)

```
\polimithesissetup{ logo-file = mylogo, watermark = false }
```

Key	Default	Effect
`logo-file`	`logopolimi3i`	Logo on the cover / first page (the Scuola 3I logo is shipped).
`watermark`	`true`	The “raggiera” in the background (cover, first page, blank pages).

6. Front-matter blocks

These come from the core class and work in every format; PoliMi only changes their look.

Block	Options	classical	article / summary
`abstractthesis`	`language`, `keywords`, `title`	Unnumbered chapter in the contents; keywords as a bold line	Unnumbered section; keywords in a blue box
`originality`	`language`, `title`	Unnumbered chapter	Unnumbered section
`acknowledgements`	`language`, `title`	Unnumbered chapter, upright text	Unnumbered section
`\dedication`	`language`	Own page, right-aligned italic	same

`language=` switches the heading and the "Keywords" label to that language (e.g. "Abstract in lingua italiana", "Parole chiave") and its hyphenation. Without it, the main language is used.

7. The formats in detail

7.1. classical

As in `Thesis_classical_format`:

- **Page**: A4, 12pt, two-sided; inner margin 3 cm, outer 2 cm, top/bottom 2.5 cm; 1.5 line spacing; no paragraph indent, space between paragraphs.
- **Cover** (`\frontpage`): Scuola 3I logo, title in blue, "Master's Thesis in *degree*", "Author: *name*", and at the bottom Student ID, Advisor, Co-advisors, Academic Year. Raggiera at the top right.
- **Chapters**: big blue number and bar, "**1**| Chapter title". Starred chapters (`\chapter*`, e.g. for the Introduction) are added to the table of contents and to the running header.
- **Running header**: page number outside, "1| Chapter title" inside, no rules. Chapter opening pages show only the page number, on the right.
- **Sections**: blue, numbered "1.1.", "1.1.1.".
- **Blank pages** left before a new chapter carry the raggiera; `\blankpage` inserts one by hand.
- **Front matter** in roman numbers (`\frontmatter`), main text in arabic (`\mainmatter`).

Typical order: `\frontpage`, `\frontmatter`, English and Italian abstracts, `\tableofcontents`, `\mainmatter`, chapters, bibliography, `\appendix`, lists of figures and tables, acknowledgements.

7.2. article

As in `Thesis_article_format`:

- **Page**: A4, 11pt, 2 cm margins, no indent; page number centred at the bottom, no header.
- **Page 1** (`\frontpage`): no separate cover. A header block with logo, title, "Master's Thesis in *degree*", "*author*, *student ID*", a rule, then a narrow column (Advisor, Co-advisors, Academic Year) next to the **abstract**, and the **keywords** in a blue box. The text starts right after it on the same page.

- **Sections** instead of chapters: blue, “1.”, “1.1.”.

```
\documentclass[university=polimi, master, format=article]{academicthesis}
\thesissetup{
  ...,
  abstract = {The English abstract, printed on page 1.},
  keywords = {here, the keywords},
}
\begin{document}
\frontpage
\section{Introduction}
...
\appendix
\section{Appendix A}
\clearpage
\begin{abstractthesis}[language=italian, keywords={qui, le parole chiave}]
  Abstract in lingua italiana.
\end{abstractthesis}
\begin{acknowledgements} ... \end{acknowledgements}
\end{document}
```

The `article` class has no chapters. A `\chapter` still compiles (it is set as a section, with one warning) so that a classical document can be switched over, but use `\section` in this format.

7.3. `summary` (Executive Summary)

As in `Executive_Summary`:

- **Page**: A4, 11pt, two columns 30 pt apart, 2 cm side margins, no indent.
- **Page 1** (`\frontpage`): a full-width block across both columns with logo, “Executive Summary of the Thesis”, title, “Master’s Thesis in *degree*”, Author, Advisor, Co-advisor(s), Academic Year and a rule; the two columns start right after it. Raggiera at the top right.
- **Following pages**: a blue box at the top with “Executive summary” on the left and the author on the right; page number centred at the bottom.
- **Sections**: blue, with tighter spacing than the article format.

The student ID is not printed, so it is not required here. As in the template, the summary normally ends with a short essential bibliography.

8. Colours, theorems and other commands

- `bluepoli` (CMYK 0.4, 0.1, 0, 0.4) is defined in every format; in `article` and `summary` it is also available as `bluePoli`, the name used by the official templates. Use it freely, e.g. `\rowcolor{bluepoli!40}` (needs `colortbl`).
- `theorem` and `proposition` environments are ready, with blue bold heads and italic text, numbered by chapter (classical) or section (article, summary). Further `\newtheorem`s use the same style.
- Figure and table caption labels are blue (bold in classical).
- `\blankpage` (classical): an empty page with the raggiera, not counted.

9. What the package loads, and what is up to you

Only what the layout needs is loaded, to avoid clashes:

Format	Packages loaded
classical	titlesec, caption, setspace, eso-pic, transparent, amsthm
article	titlesec, caption, eso-pic, transparent, amsthm, tcolorbox
summary	titlesec, caption, eso-pic, transparent, amsthm, tcolorbox

The core class adds graphicx, geometry, fancyhdr, xcolor, amsmath, lettrine, babel and biblatex.

Left to you, as in the "add your packages" section of the official templates: `hyperref` / `cleveref` (the layout works with or without them), `amssymb`, `bm`, `float`, `colortbl`, `subfig`, `tikz`, `algorithm`, `enumitem`, and the **bibliography**: the package sets no style, so configure it in your preamble, e.g. `biblatex-options={style=numeric-comp}` as a class option, then `\addbibresource{...}` and `\printbibliography`.

10. Languages and labels

The labels come from string files, so a thesis in Italian gets Italian labels and one in English gets English labels.

String	English	Italian
Author	Author	Candidato / Candidata
Student ID	Student ID	Matricola
Advisor(s)	Advisor(s)	Relatore / Relatori
Co-advisor(s)	Co-advisor(s)	Correlatore / Correlatori
Thesis type	Bachelor's / Master's Thesis in	Tesi di Laurea Triennale / Magistrale in
Abstract heading	Abstract	Abstract in lingua italiana
Keywords	Keywords	Parole chiave
Summary line	Executive Summary of the Thesis	Executive Summary della Tesi
Theorem, Proposition	Theorem, Proposition	Teorema, Proposizione

To change a wording, edit `academicthesis-polimi-<language>.ldf` (or add a file for a new language with the same keys).

11. Common errors

Message	Cause	Fix
Unknown format ‘classic’ for university ‘polimi’	Typo in `format=`	Use `classical`, `article` or `summary`.
Missing required thesis metadata: ‘abstract’	`format=article` without the abstract	Add `abstract={...}` (and `keywords={...}`) to `\thesissetup`.
Missing required thesis metadata: ‘idnumber’	No student ID (classical, article)	Add `idnumber={...}`.
No front page registered for university ‘polimi’, degree ‘phd’	PhD is not available yet	Use `bachelor` or `master`.
Warning: The base class ‘article’ has no chapters	`\chapter` in article or summary	Use `\section`.
Warning: ‘french’ … was not loaded by babel	A block with `language=french`	Add `other-languages={french}`.

Compile with pdfLaTeX (the default). In Texifier/Texpad, set the typesetter to TeX Live (pdfLaTeX), not the built-in engine.

12. For maintainers: how the package is organised

File	Role
`academicthesis-polimi.cfg`	Read by the class *before* it loads its base class. Declares the formats (`\DeclareThesisFormats`), the base class and default options of each format, no second page, and English + Italian as extra languages.
`academicthesis-polimi.sty`	`\polimithesissetup`, required metadata, loads the packages of the chosen format, then reads that format’s `.def`.
`academicthesis-polimi-classical.def`	Classical layout.
`academicthesis-polimi-article.def`	Article layout.
`academicthesis-polimi-summary.def`	Executive summary layout.
`academicthesis-polimi-english.ldf`, `-italian.ldf`	PoliMi wording of the core labels, plus PoliMi-only strings.
`logopolimi3i.pdf`, `raggierapolimi.pdf`	Logo and background, converted from the template’s EPS.
`academicthesis-polimi-doc.tex`	This manual.

Rules to keep in mind when editing:

- The `.def` files are written in **normal LaTeX syntax** (they are read with document catcodes), so spaces and blank lines behave as usual. The `.sty` and `.cfg` use LaTeX3 syntax, where spaces are ignored (write `~`).
- **Never load a package from a `.def` file**: when that package ends, LaTeX switches LaTeX3 syntax back on and every following space is lost. Add the package to the format's list in the `.sty` instead.
- Each `.def` registers its own front page with `\DeclareThesisFrontPage` (for `bachelor` and `master`) and changes core pieces (page styles, acknowledgements, keywords) with `\AddToHook` on the hook `class/academicthesis/after`, because the core defines them after loading this package.
- Metadata is read with the core accessors `\thesismeta{key}`, `\IfThesisMetaEmptyTF`, `\IfThesisListEmptyTF`, `\IfThesisDegreeTF`, and labels with `\thesisstring{Key}`.

Adding a format: add it to `\DeclareThesisFormats` and to the base-class switch in the `.cfg`, list its packages in the `.sty`, and write `academicthesis-polimi-<format>.def`. **Adding PhD**: register a `phd` front page with `\DeclareThesisFrontPage` in each `.def` that supports it.